3.3. Platforms

3.3.1. Imaging

The Imaging Platform provides functionality to support:

- inter-process communication (IPC)
- display management
- network filters (software firewall)
- antivirus

Some aspects of this software are described in section 2.4.8 of this document.

3.4. Libraries

3.4.1. FFTW

The FFTW package is used by Viz, MR Recon, PSS. The software provides a Fast Fourier Transform mathematics library. The FFTW software is further described in the Recon Libraries SOUPRS [36] and FFTW VIZ SOUPRS [33].

3.4.2. Boost

All C++ software benefits from a variety of language enhancements not yet supported in the compiler versions used in product.

This design specification identifies Boost as the means to provide enhancements for more effective C++ code. Boost's capabilities range from general-purpose libraries like the smart pointer library, to CRC libraries, to operating system abstractions.

4. STRUCTURE

The Software functionality is distributed across multiple subsystems. The sections below describe the primary components of the Software subsystems, their responsibilities, and the primary paths of communication with other subsystems.

4.1. Host Software

Figure 4-1 shows the organization of Host within the Software.

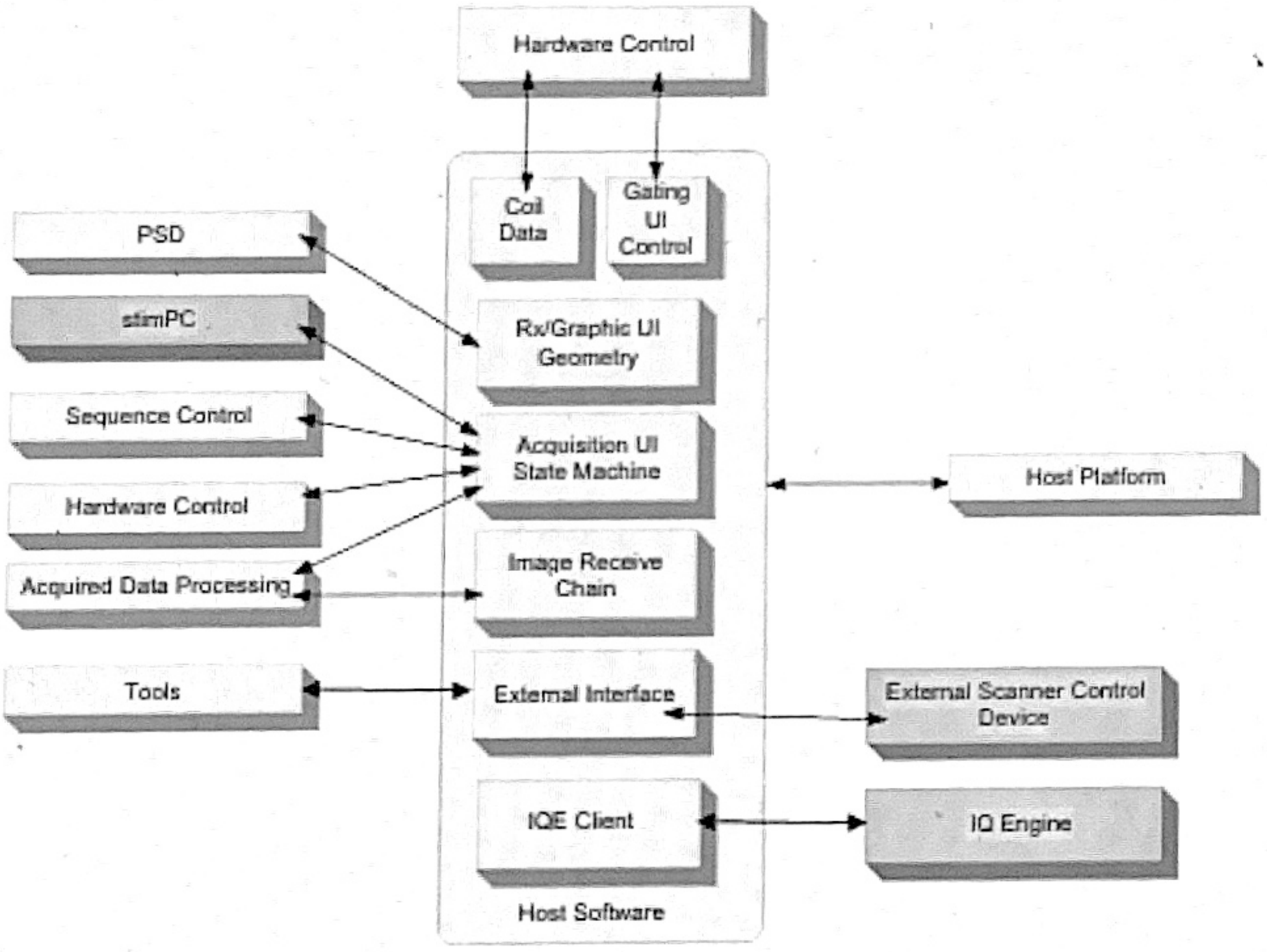


Figure 4-1: Host Software Logical Organization

The detailed design for Host is covered in the Host Software SSDD [35].

4.1.1. Rx UI

This component is responsible for displaying & controlling the UI for MR prescriptions. It communicates with PSD to validate user input and other parameters for the selected Sequence. It also provides the Coil Plug selection interface and manages automatic coil selection in conjunction with the Coil Data Management component.

4.1.2. Graphic Rx UI

This component enables graphical way to prescribe scanning range parameters. It allows user to prescribe ROI for acquisition, sat bands, shimming, tracking etc.

4.1.3. Geometry

This component converts data between the fundamental coordinate systems used in the Scanner: Mechanical, Table and Patient.

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- Patient coordinates are associated with subjects being scanned. They are used by Operators and Reading Physicians to locate and measure subject anatomic structures and motion vectors. The Software applies Patient coordinates when it accepts input during prescriptions and when it annotates images.
- Mechanical coordinates are associated with the Scanner hardware that controls the MR Physics during acquisition and to the MR Physics itself. They are used by service engineers to locate and measure the magnet, gradients, and coils. The Scanner applies Mechanical coordinates when calculating and controlling fields and vectors relevant to the spatial location of acquisition volumes.
- Table coordinates are associated with Patient Handling components and their movements relative to the Scanner hardware. They are used by service engineers to locate and measure the position and speed of the Table, Cradle, and LPCA. The Scanner applies Table coordinates when locating and moving Patient Handling hardware.

4.1.4. Acquisition UI

This component is responsible for displaying & controlling acquisition UI. This is used by real time, Prescan and spectro application. This communicates user input to AGP & VRE and displays response.

4.1.5. State Machine

This component is responsible for managing the state of the Scanner. The Scanner goes through different states from prescription to scan done. This component controls scanner UI based on the state. It also enables automatic/manual change of state based on user choice.

4.1.6. Image Receive Chain

This component is responsible for generating, displaying & installing images. Based on user prescription it initially generates image header. Then it combines images header with pixel data from MR Recon. Finally it generates DICOM image using translator. This image is then displayed automatically in image viewer and is submitted for installation.

4.1.7. External Interface

This component (EXSI) provides a programmatic interface to the scanner, allowing remote control capability, used by focused ultrasound. It also provides a scripting capability, used by system tools to calibrate the Scanner.

4.1.8. Coil Data Management

This component (CDM) controls access to the coil data, processes coil codes, provides coil data to consumers, as well as computes the settings necessary to operate the Scanner's connected coils. CDM can operate in one of two modes: static and dynamic. In static mode, an Operator inputs the positions of floating Coils, and the system applies predefined combinations of Coil Elements. In dynamic mode, the system acquires low resolution 3D data from all available Coil Elements, and then dynamically identifies the combinations of Coil Elements that

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provide appropriate signal. Additionally, the JSTD curve data associated with the SAR calculations are managed within the coil database and protected as described in Section 8.1.

4.1.9. Gating UI

This component provides the interface and control functions to set up the cardiac and respiratory gating subsystem as well as the waveform display.

4.1.10. IQE Client

This component provides an interface through which the Scanner communicates with the Image Quality Engine (IQ Engine). The IQE Client provides a UI by which Scanner users command IQ Engine operations, a tool to access encrypted log information to be provided to the IQ Engine, and an API that provides data to and accepts results from the IQ Engine.

The IQ Engine is a separately sold product that generates specialized Images, typically applying DL Reconstruction for Scanners that are not licensed for DL Reconstruction Options. The architectural design of the IQ Engine is covered in the IQ Engine SWADD [47].

4.2. Host Platform

Figure 4-2 shows the major components within Host Platform as well as the primary communication connections with other subsystems.

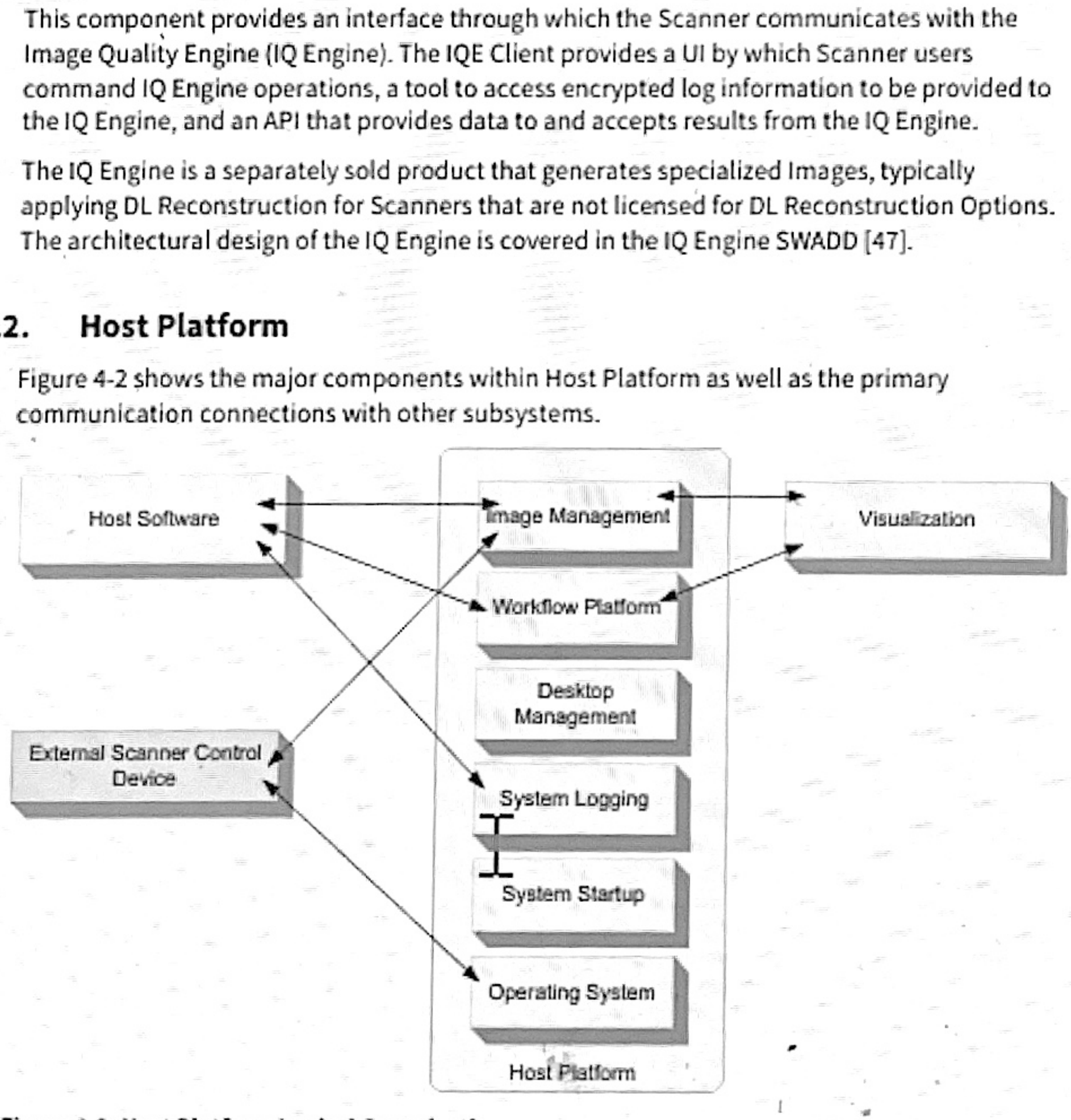


Figure 4-2: Host Platform Logical Organization

The paragraphs below further describe each of the Host Platform components.

4.2.1. Image Management Platform

This component provides the ability to locate exams, select processing applications and initiate processing sessions as well as move the exam data to/from network destinations and/or local storage media.

4.2.2. Workflow Platform

This component is responsible for managing sessions & tasks. Session management involves starting/closing several sessions. Task manager provides the high level control for the scan and post process tasks within a scan session. It also owns modality work list and protocol management.

4.2.3. Desktop Management

This component provides overall control of user desktop, managing the visibility of various UI component groups.

4.2.4. System Logging

This component provides the libraries necessary for message handling as well as the components necessary to provide message display. It also displays the Scanner status, such as networking status, archive status, recon status, etc.

4.2.5. System Startup

This component is responsible to startup the Host processes in the proper order including synchronization with external subsystems.

4.2.6. Operating System

This component comprises the operating system kernel as well as all operating system services, libraries, and applications provided by the operating system SOUP.

The detailed design for Host Platform is covered in Host Platform SSDD [38].

4.3. Visualization Applications

The Visualization Applications subsystem (aka Visualization, aka Viz) is a set of post-processing applications that load images and other data, analyze the inputs, and either display the results on the console or persist the results in the Image DB. Applications are typically run as review sessions using data from closed exams (aka "Session Applications"), but several applications provide interfaces compatible with workflow management allowing the functionality to be prescribed and invoked within a scanning session (aka "Inline Applications"). Inline Applications replace live user interactions with prescribed or default settings. Session Applications and Inline Applications provide identical results often by utilizing common shared libraries.

Inline Applications may be prescribed to execute automatically or manually. Automatic processing relies on the workflow platform to invoke the Application when the prescribed

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data are ready to be processed and does not allow user interactions after invocation. Manual processing relies on the workflow platform to determine when the data needed for the Application is ready, but requires the user to invoke the Application and allows user interaction.

The typical visualization has the following data-flow.

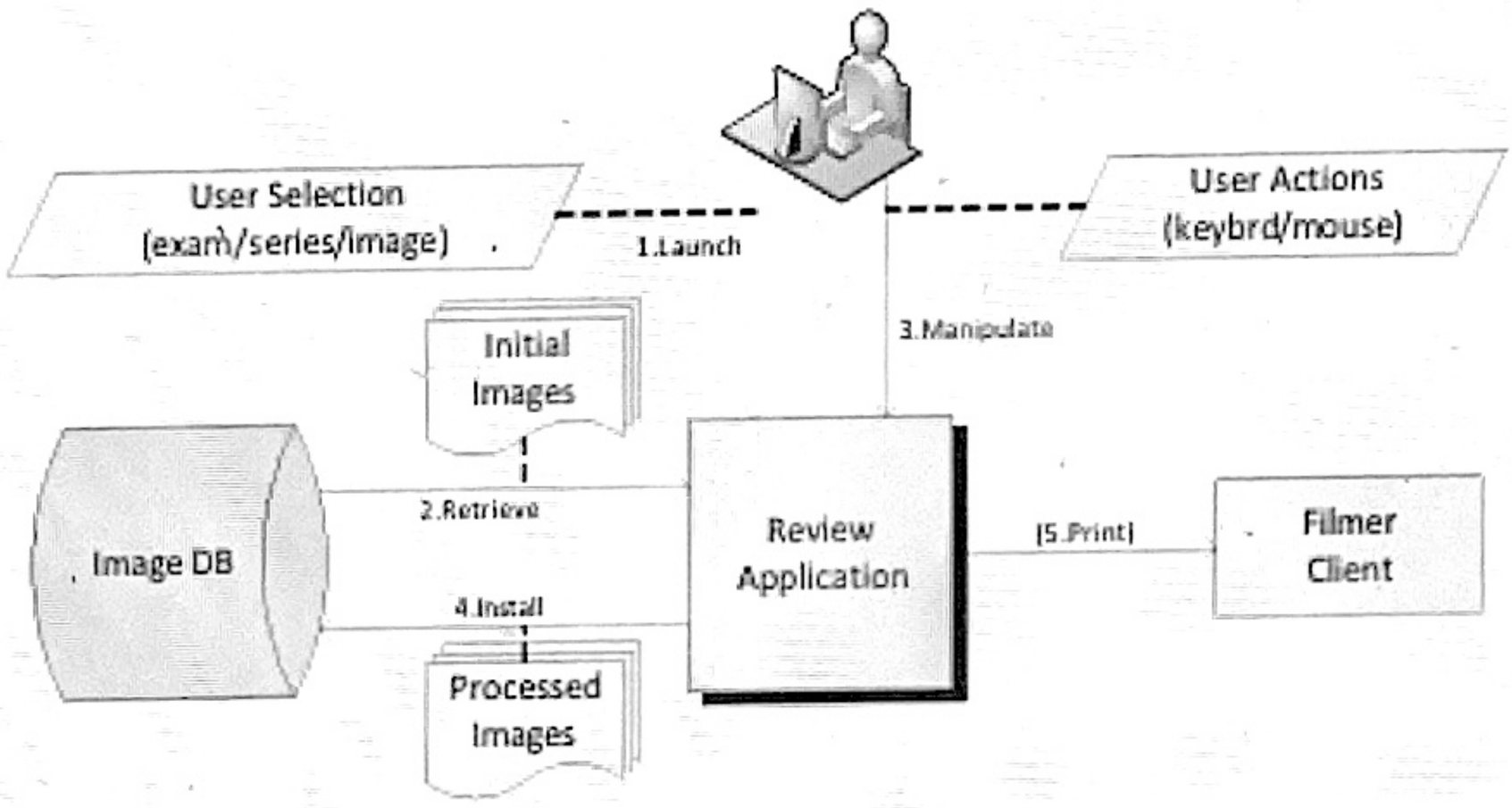


Figure 4-3: Review Session Application Data Flow

For inline processing, the dataflow is shown below.

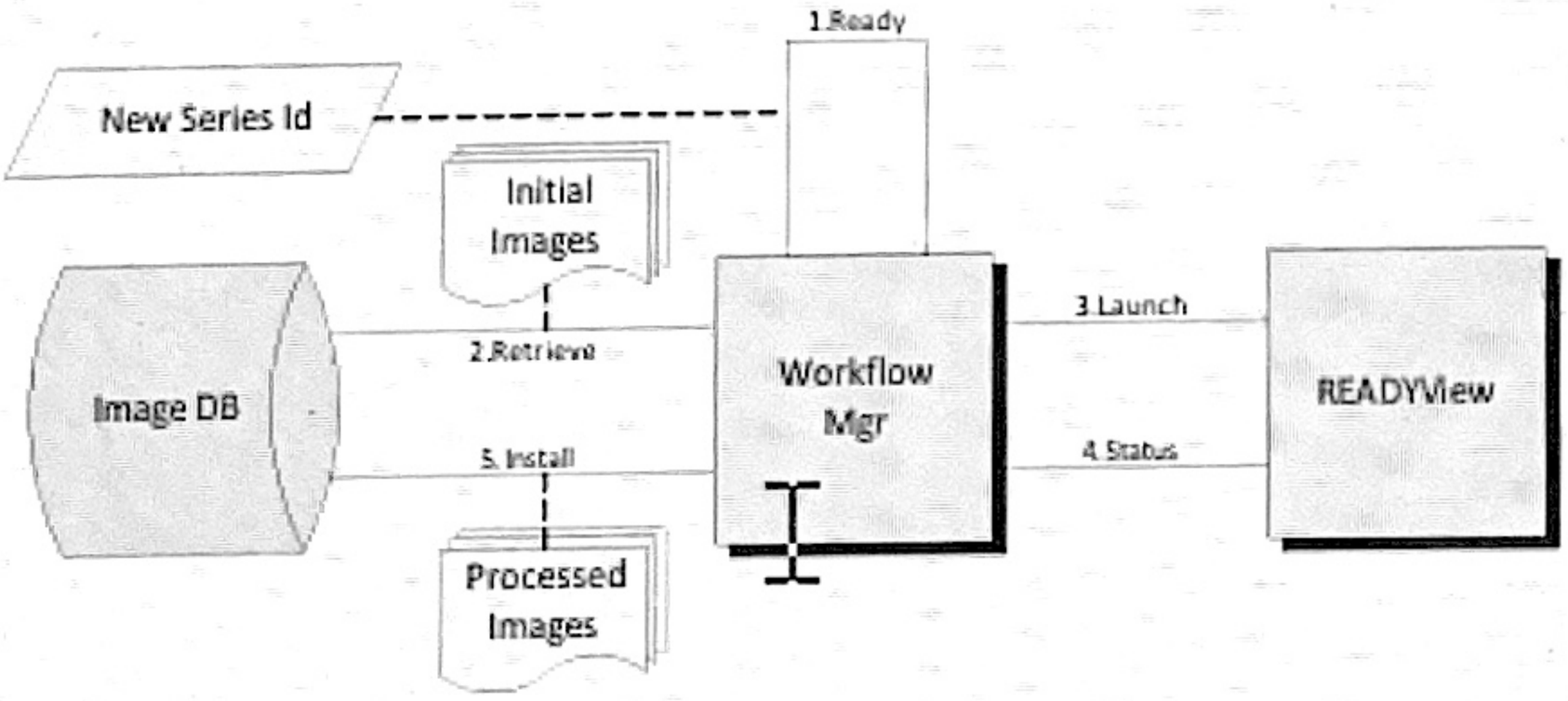


Figure 4-4: Scan Session Application Data Flow

The detailed design for Viz is covered in the MR Visualization SSDD [39].

4.4. MR Reconstruction

The MR Reconstruction subsystem (aka MR Recon) is responsible for managing the arrival of acquired MR data frames from the hardware Receive Chain, processing those frames, and

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generating images, spectrographs (in MR spectroscopy), scanner hardware calibrations, and real-time feedback to the PSD.

MR Recon is also responsible for managing Prescription Inference Engines, which generate scan parameters for Prescription, managing access to Raw Data, and managing the VRE start up, reset, and diagnostics functions.

MR Recon consists of several processes, including:

- Acquisition Data Server (ADS)
- Prescan Server
- Data Processing Server (DP)
- Real-Time Processing Server (RTP)
- Image Recon
- Anatomy Recognition Service
- Raw Data Server (RDS)

ADS is responsible for routing received data frames. Prescan Server generates scanner hardware calibration data. DP, RTP, and Image Recon create images and spectographs. Anatomy Recognition Service generates prescription parameters. RDS provides external access to acquired data and control packets.

The detailed design for MR Recon is covered in the MR Reconstruction SSDD [40]

4.4.1. Acquisition

The Acquisition functionality of MR Recon runs on the VRE, which includes the hardware interface to the Receive Chain. MR Recon takes advantage of the multi-core computer to provide good clinical performance.

The hardware interface to the Receive Chain uses the integrated Volume Reconstruction Receive Filter (iVRF), which is a PCIe extender card connecting to the SSC subsystem where the filter modules reside. The iVRF collects multi-channel MRI signal data from the Receive Chain and utilize DMA to provide frame and control data into the ICN memory. A Linux driver on the ICN manages the device and a API library is used to interface with the driver.

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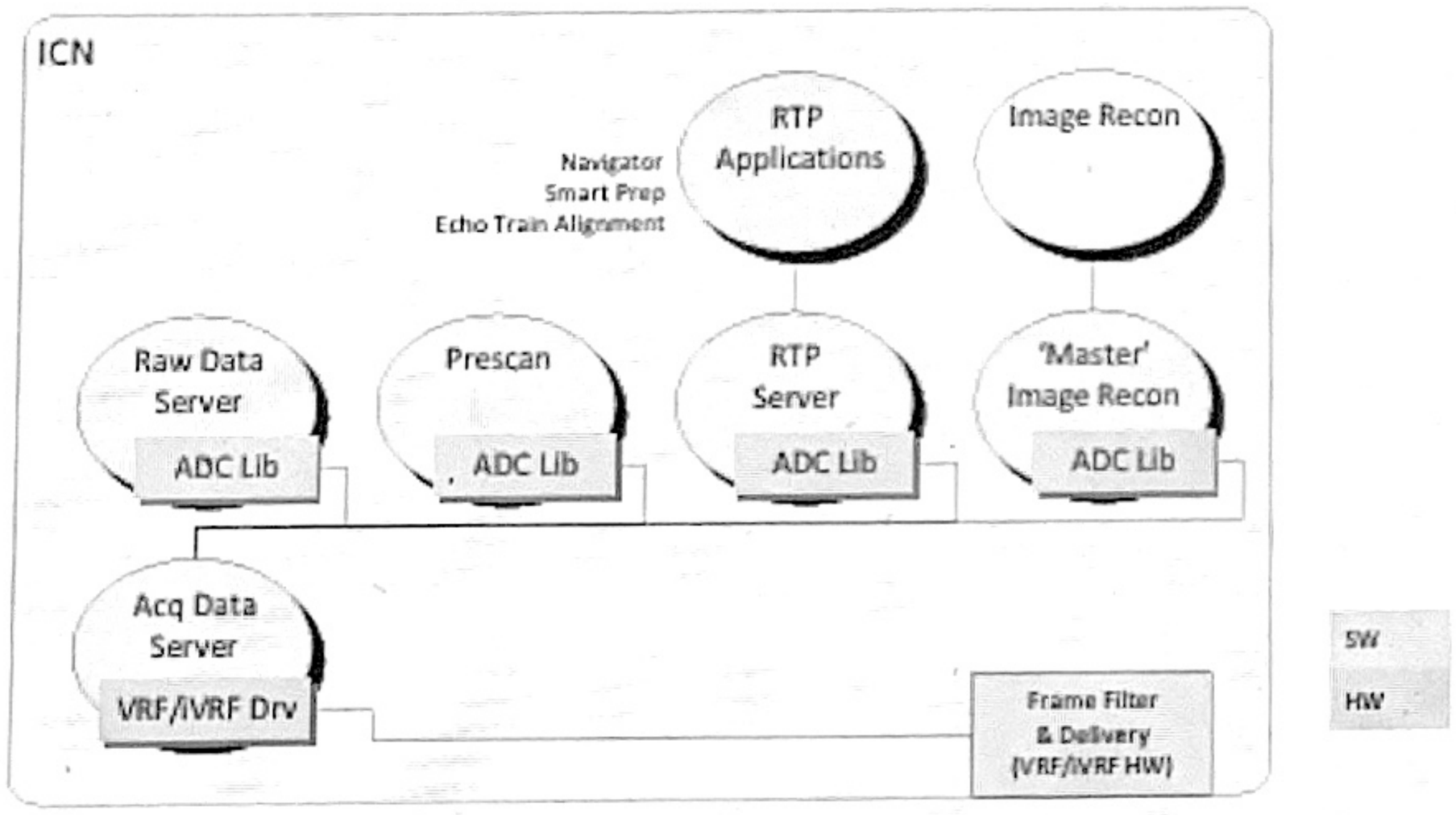


Figure 4-5: Acquisition Clients

Data frames arrive from the Receive Chain HW at a fixed rate (±250 KHz). The iVRF hardware component filters frame data to the prescribed MR bandwidth and the prescribed number of points in the frequency-encode direction. Digital gain (actually attenuation) adjusts the dynamic range of the received data so it is either 32-bit or 16-bit complex data. (Because complex number representations include both real and imaginary parts, each element occupies 8 bytes or 4 bytes, respectively.) The iVRF inserts the received data frames directly (DMA) into the ICN memory. The iVRF also receives DAB packets from PSD and inserts them into the ICN memory.

The Acquisition Data Server (ADS) process manages the iVRF hardware. Data is received in the order the PSD plays it out. The ADS also provides an interface allowing the other MR Recon processes to register as listeners for received data frames and DAB packets. The ADS delivers those to registered listeners as they arrive. The Acquisition Data Client (ADC) library is used by each of the other processes to facilitate the inter-process communication for client registration, frame delivery, and DAB packet delivery. This keeps the application code in the "worker" processes independent of the details of acquisition. The ADS and frame delivery is the critical real-time processing within MR Recon. The ADS uses the API associated with the underlying hardware to abstract away the hardware specific details. The ADS in turn abstracts this to all clients.

4.4.2. Image Generation

The Image Generation functionality (aka Orchestra) is the best-known portion within MR Recon. Orchestra is organized as shown in Figure 4-6.

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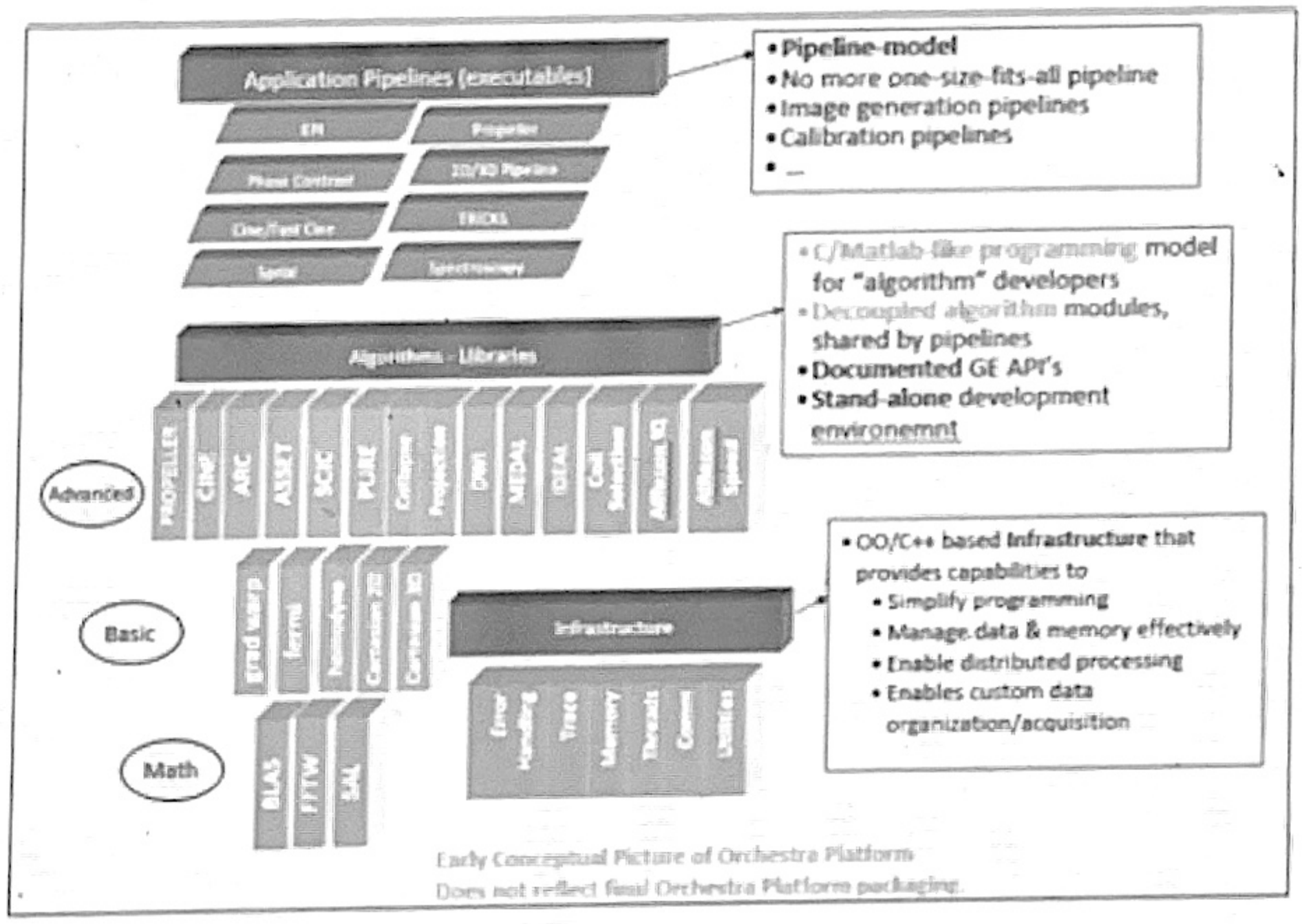


Figure 4-6: Orchestra Organization

An Orchestra pipeline is assembled from modules driven by parameters that specify how to generate images, including types and number and order of outputs, identification policies, fundamental algorithms (e.g. fft, DL), and enhancements (e.g. filters). Each assembled pipeline is executed as a process that prepares the scan, allocates the storage needed for the incoming data frames, accumulates incoming raw data until there is a sufficient dataset to begin processing, and processes the raw data. Orchestra runs a variety of algorithms that transform the raw data into MR images and MR spectrographs. Orchestra's outputs are sent to Host. Orchestra processing is designed for distributed computing, taking advantage of the multi-core ICN computer. For pipelines with intensive processing, Orchestra may take several minutes to generate its outputs, soonce the acquisition of a pipeline is completed, Orchestra allows the next pipeline to prepare and acquire, provided enough storage is available.

The Acquisition Handler within a pipeline interfaces with ADS to access Raw Data. It uses the control packets in the raw data to monitor the acquisition status. The Acquisition Handler keeps processed data in a Runtime Database organized hierarchically based on attributes associated with the processed data, including the Scan, Acquisition, Slice, Echo, Channel, and Context. Each application defines its organization. Processing generally starts at the bottom with acquired data and flows up the tree as depicted in Figure 4-7.

Data handling in the Runtime Database is highly efficient. Zero-copy hand-offs are used with the Contexts passing data. Bulk data may be stored to disk as needed and brought back into memory when needed. The Runtime Database is a powerful enabler for Orchestra, high

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performant while providing excellent encapsulation of complex multi-dimensional data to the benefit of simpler algorithms and sections within each pipeline.

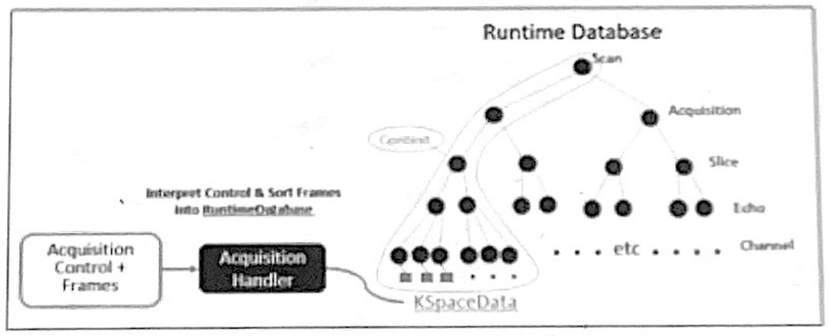


Figure 4-7: Runtime Database Organization

Each module of a pipeline transforms data from one form to another. Each module is invoked when input data is available. The module obtains its data from the Runtime Database via a Context, performs its processing on its data, and stores its results to the Runtime Database associated with a related Context. The Context recognizes the Runtime Database hierarchy, encapsulating access of data that is input and output for the modules.

4.4.3. Prescan

The Prescan functionality of MR Recon tunes the scanner for the Patient by running a series of "passes" to establish the Center Frequency, Transmit Gain, Receiver Gain (analog and digital), shim settings, and Channel Noise weights appropriate for the Patient. Prescan also optionally runs Coil calibration, ASSET/PURE calibration, spectro Prescan, EPI reference, and FSE phase correction passes. Prescan may be run in Manual mode or in Auto mode, which requires no operator intervention. It accepts the received data frames and performs pass-specific computations to determine the appropriate tuning values. In Manual mode, Prescan displays the Manual Prescan UI, at which the operator sets the tuning values while viewing the acquired frame data. In both Manual and Auto modes, Prescan sends the final tuning values to the transmit chain and receive chain hardware. Prescan also provides the final tuning values to Host Prescan and Image Generation.

4.4.4. Anatomy Recognition

The Anatomy Recognition functionality of MR Recon processes generated images. Typically these are localizer images, but other images are used for certain anatomies. For each anatomy (e.g. head, knee, spine, heart), the Technologist normally looks for fiducial markers in the patient's anatomy to orient the scan planes of the prescribed slices. The Technologist observes images in Graphic Rx to correctly align scan planes for each patient and for various scanning tasks on that patient's anatomy.

The Anatomy Recognition services and the deep learning trained neural inference engine replicate those manual operations to determine the appropriate slice plane orientation from

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the localiizer and additional images. The protocol or the operator specify the desired anatomical fiducial marker. That, along with the patient's reference images, enables calculation of a recommened scan plane. The recommended scan plane is provided to the UI and displayed in Graphic Rx, and the Technologist may use it or make adjustments as needed.

4.4.5. Real-Time Processing

The Real-Time Processing functionality of MR Recon consists of an RTP Server and one of several RTP Applications. These applications run specialized algorithms on acquired frame data and provide feedback to the PSD while the scan is running. The PSD takes action based on that feedback. Low latency is critical for RTP processing and feedback.

4.4.6. Raw Data Access

The Raw Data Access functionality of MR Recon provides external access to acquired data and control packets for research sites, and is not available when the scanner is in clinical mode. The RDS package includes a Raw Data Client library for the researcher to include in their remote process. This library abstracts the inter-process communications and the details of the wire-formats.

4.5. Pulse Sequence Software

The Pulse Sequence Software (aka PSS) subsystem, realized as a set of Pulse Sequence Descriptions (PSDs), accomplishes two primary tasks. The first is to utilize the inputs from the user to configure the gradient, RF, and control subsystems to conduct a particular MR physics experiment. This physics experiment realizes the user's request while also meeting the constraints of MR physics, patient safety, hardware, and software. In serving this function, the PSD can be seen as a model that receives and validates inputs and produces output parameters for display to the user through the user interface subsystem or for communication to other subsystems. Since this model evaluation cycle is driven by user inputs, response times on the order of human interaction speeds (tens to hundreds of milliseconds) are acceptable.

The second task realized by PSS is to feed waveform and control data to the relevant hardware and software subsystems to excite and manipulate the spins to acquire the images based on the user specified parameters. Due to the time scale of the MR physics involved, reliable performance with control on the order of ones to tens of microseconds is necessary.

Since the performance needs of these two tasks vary so greatly, the subsystem and PSDs are designed with separate computational sections and separate executable code. The first process (referred to as the "Host PSD") runs on a standard computer operating system. The second process (referred to as the "Target PSD") runs on a real-time operating system with predictable, reliable scheduling and performance to meet the needs of the runtime PSD.

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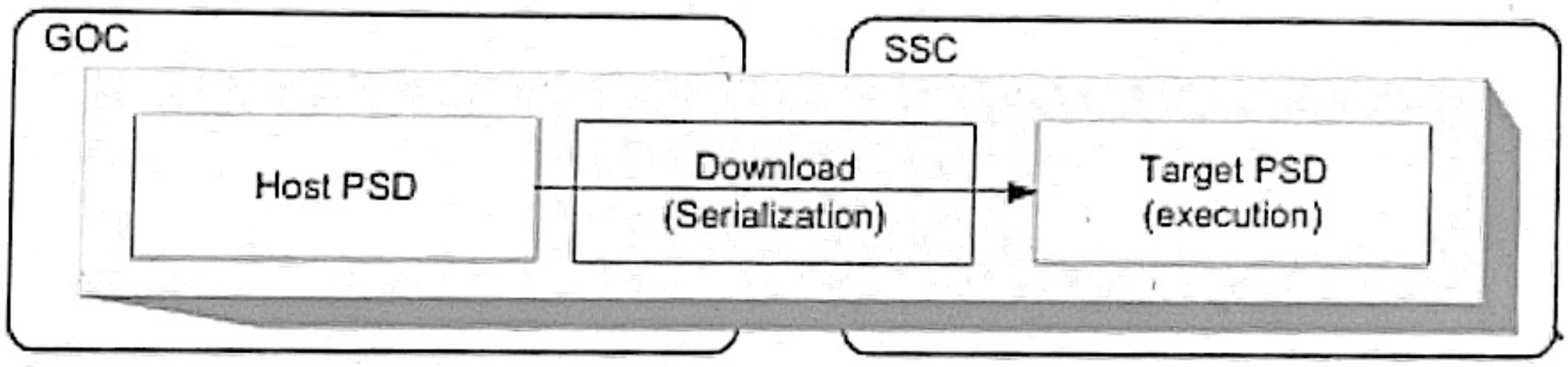


Figure 4-8: PSS Deployment Model

Each Host PSD is a separate executable that implements one or more applications. Multiple concurrent instances of the same Host PSD can exist at the same time to allow multiple series to use the same Host PSD executable to setup and acquire different images. Each Target PSD may contain multiple waveform descriptions that can be activated individually to accomplish unique MR experiments. Each individually accessible experiment is called an Entry Point. The two PSD processes share information through a parameterized serialization/de-serialization process called download.

Detailed design of PSS is described in the Pulse Sequence SSDD [42].

4.6. MR Firmware

The MR Firmware (aka MR FW) subsystem is decomposed into the following components.

4.6.1. Pulse Sequence Interface

The Pulse Sequence Interface (PSI) provides access to the PSS, allowing sequence information to be loaded into the sequencing state machine. This is achieved by building dynamically linkable object modules for each of the PSDs. As part of the scan download process, the appropriate module is loaded and linked into PSI. The PSD is then able to access the functional interfaces to play out waveforms and control hardware during acquisition.

Internally the PSI is partitioned into a sequencer component, rotation and eddy current correction component, and hardware interface component. The hardware interfaces are used to play out specific waveforms and to change the operating mode or set up of hardware components. For example, for a particular calibration, the pulse sequence may need to change the mode of the RF amplification from head output to test port output.

The PSI accepts external gating inputs during sequence play out. These gating inputs allow the sequence play out to be synchronized to external events, such as the patient's respiratory cycle, or cardiac cycle.

The detailed design for PSI is covered in the Firmware SSDD [42].

4.6.2. Scan Control Processor

The Scan Control Processor (SCP) maintains and sequences the scan state of the Scanner, from the time an application is downloaded, until it is scanned and unloaded. SCP interfaces with MR Recon, PSD, PSI, Host, and HW to communicate state information. These subsystems also communicate to SCP to perform hardware setup, download, prep, start, pause, and stop

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scan events. The SCP state transitions ensure that the hardware is fully setup, and without fault conditions before a scan can be started or resumed. It also handles stopping or pausing the scan on request or if hardware is faulted.

The detailed design for SCP is covered in the Firmware SSDD [42].

4.6.3. Receive Chain Software

The Receive Chain Software section consists of the coil input interfaces to the Scanner and the RF signal components between the coil input interfaces and the digitized signal that is offset corrected, demodulated, filtered, and decimated from the output of the Receiver Analog to Digital Converter. The receive chain software resides, as firmware, in the iVRF or ICE modules and in the RF Control section in the receiver input multiplexer.

4.6.4. Transmit Chain Software

The primary functions of the transmit chain are to generate the whole body B1 field needed for MR imaging and to provide the necessary signals to support generation of the B1 fields by Head/Local Transmit coils. However, the Transmit Chain also serves as the source for reference and gating signals that are used to synchronize Scanner data acquisition functions and transmit/receive RF signal processing functions. The transmit chain also provides the means of measuring RF power being transmitted to the patient load, which is then used for system-level SAR management.

4.6.5. Gradient Driver Software

The Gradient Driver Software section consists of the gradient drivers, gradient power supply, the resonance module and the accessories that connect the gradient driver to the resonance module. There are three gradient amplifiers, one per gradient axis (X, Y, Z). Each amp converts a digital waveform, defined by the PSD, to an analog electrical current waveform. Each current waveform drives corresponding gradient coils, which are packaged together as the resonance module. The gradient coils convert the current waveforms into magnetic fields, which are used to frequency encode a region of interest in the specimen for an MRI experiment. The gradient driver software resides, as firmware, in the control side of the gradient driver to provide the management and interpretation of gradient data and control commands from the communication links to the driver hardware. Some systems have gradient coils that support dual gradient modes which provide a smaller FOV with higher performance gradient mode and a larger FOV at a lower performance gradient mode. These systems have additional controls to select the appropriate gradient mode for the acquisition.

4.6.6. Cooling and Cabinet Monitor

For products with Heat Exchanger (HEC) hardware and no Cabinet Monitor hardware, the Cooling software provides an interface between Host and HEC. This software is responsible for collecting the current state of HEC and using it to determine if scanning is allowed and for logging performance data that can be collected and monitored by the back office.

For products with Cabinet Monitor hardware, the Cabinet Monitor software provides the same interface as the Cooling software, but manages a variety of cooling systems. Additionally, the

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Cabinet Monitor software monitors the performance of other system hardware, including various DC power supply voltages and fan speeds used by the Exciter, Gradient, In-room Display, and other components. If any monitored parameter falls outside its defined threshold, the Cabinet Monitor software inhibits scanning. Finally, the Cabinet Monitor software controls multiple system power switches, turning off the system power during normal Shut Down (see section 6.2.5) and turning on the system power when it is off and the power button on the GOC is pressed.

4.6.7. Scan Room Interface

The Scan Room Interface (SRI) software is responsible for responding to the user input at the magnet, cradle motion control, thermal monitoring of the scan room, scan position interface (landmark), and physiological acquisition. The communication of state events, such as scan position or landmark changes, are sent to Host for processing. Other events, such as move in/out button presses, are handled within the patient handling software. The physiological acquisition software sends waveform data to PSI for monitoring and to Host for waveform display.

SRI supports a special mode of operation for the touch-and-go components to allow the operator to identify the position of the floating GEM coils. This mode of operation does not impact the patient landmark.

4.6.8. Magnet Monitor Software

The Magnet Monitor Software (magmon) serves two purposes in the Scanner: magnet control and magnet monitoring. For magnet control, magmon uses a resistive heater to maintain a constant pressure in the magnet's cryogen tank. Magmon also has the ability to restart the cryogen compressor after some types of compressor shut downs. For magnet monitoring, magmon collects and stores an assortment of magnet sensor readings. For systems with a service contract, magmon periodically uploads some of this data to GE's online service center and will request a FE service call if system parameters stray outside of expected parameters. This remote monitoring serves to reduce the cost of Scanner maintenance and is not necessary for normal usage.

The detailed design for the Magnet Monitor Software is covered in the Magnet Monitor SWDD [46].

4.7. MR Service Software

The MR Service Software (aka Service SW, aka MR Tools) subsystem is a collection of relatively independent applications. The various categories of tools utilize similar architecture patterns as shown in Figure 4-9.These applications typically persist results in files for use by other tools or the application software.

The detailed designs for Service SW are covered in the Host Tools SSDD [44] and the System Tools SSDD [45].

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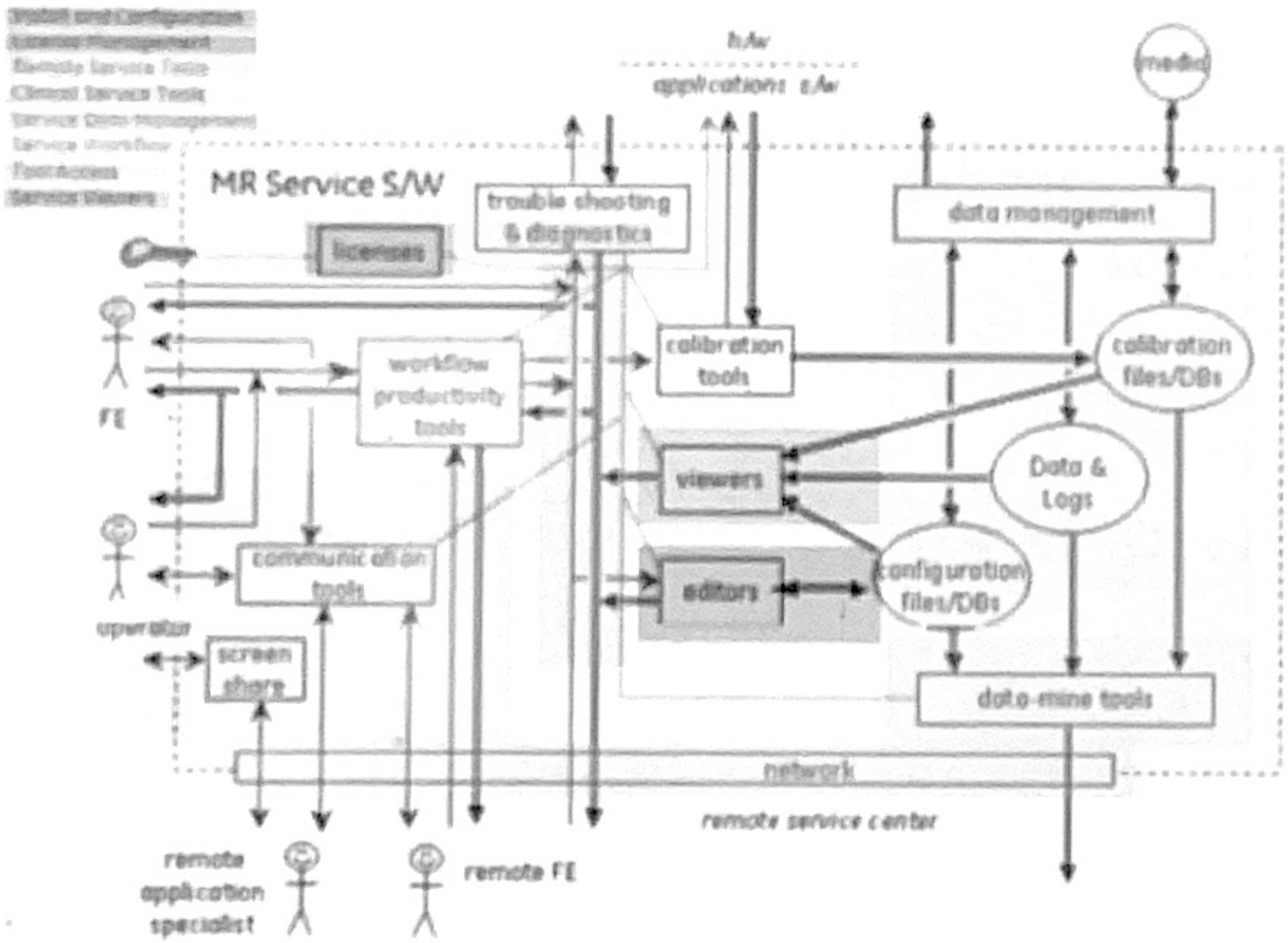


Figure 4-9: MR Service SW Block Diagram

4.7.1. Software Tools

These applications are used to perform some of the Manage Protocol uses (e.g. via Protocol Exchange), some of the Monitor System uses (e.g. via System Health), some of the Manage System uses (Install, Uninstall, Upgrade, Start Up, Shut Down, Configure Cybersecurity, Configure Licenses), some of the Qualify System uses (e.g. via Install In-Spec), and some of the Maintain System uses (e.g. via SaveInfo-RestoreInfo), and some of the Diagnose System uses (e.g. via Diagnostic Platform, FE Cockpit). These applications are available to all customers without additional licensing.

4.7.2. System Tools

These applications are used to perform some of the Qualify System uses (e.g. via Daily Quality Assurance), the Calibrate System uses, and some of the Dianose System uses (e.g. via EPI White Pixel). These applications typically provide information regarding the Scanner's signal paths to assure good image quality. These tools run Some of these applications are available only when a service license is active, and some are available without additional licensing.

These applications typically initiate scanning (via host processes) in "geservice" mode to provide performance information. This is used to reduce costs and time needed to repair the Scanner. Some results are directly presented, and other results are persisted in files to be viewed or used by other tools. System Log data is encrypted, so all service tools that read System Log records must do so using the GeSyslogUtil.

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4.7.3. Service Tools

These applications are used to perform the Monitor System uses (e.g. via Image Performance Monitoring), some of the Qualify System uses (e.g. via Installed Base In Spec), some of the Maintain System uses (e.g. via SW Auto-Download), and some of the Dianose System uses (e.g. via InSite Interactive Platform). These applications are provided to MR by GEHC Service organizations (DST, DPS).

These applications include:

- Viewers for calibration and configuration data, as well at data and logs files created by the application software.
- Data mining tools to collect and send data to the remote service center (On-Line Center) for advanced services. Some tools do this on schedules, others on remote request.
- A screen sharing tool that allows a remote application specialist to share the user interface with the operator at the site.
- Networking tools to provides basic network functions and connectivity to the remote service center.

5. INTERFACES

5.1. User Interfaces

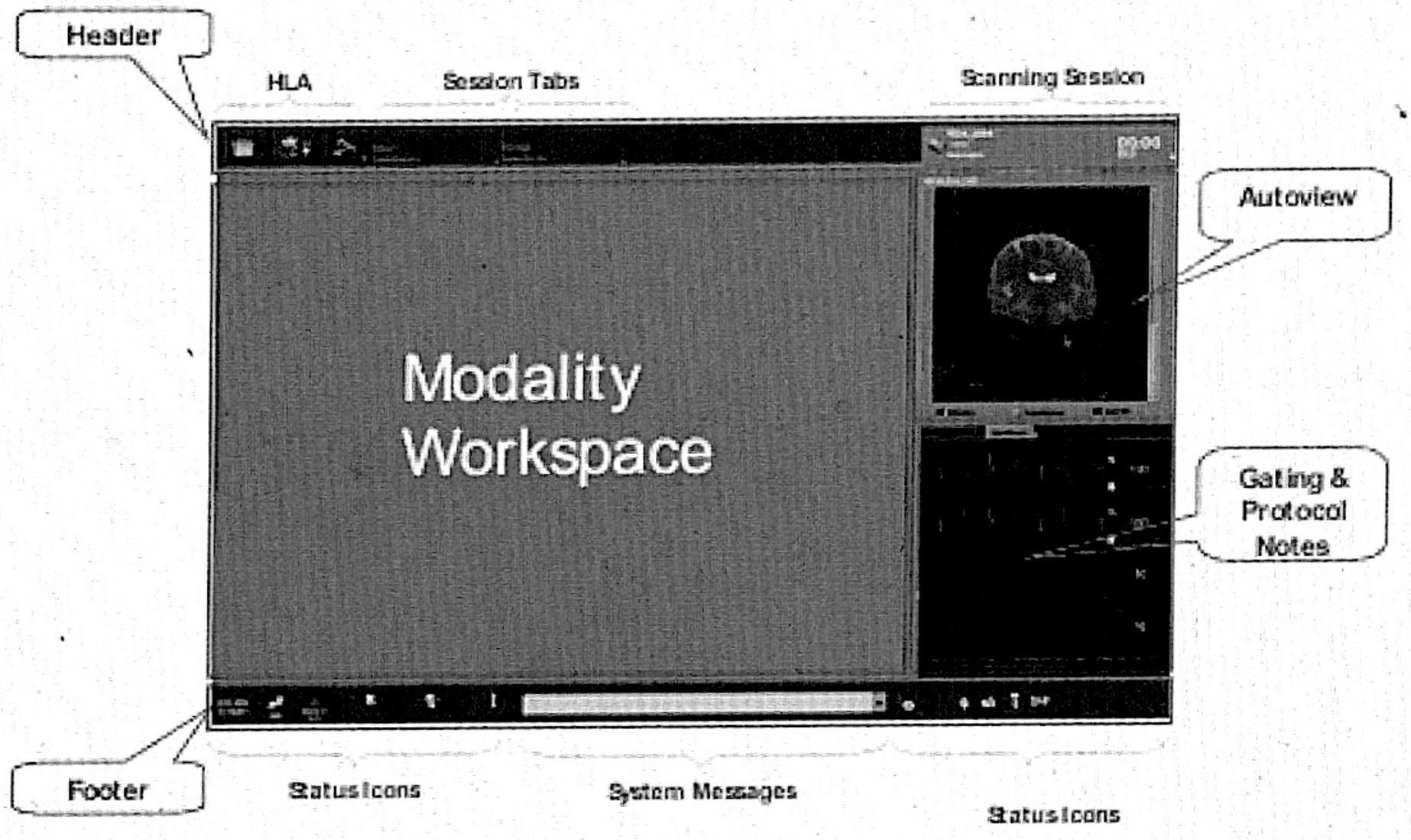
The user interface is designed to provide a common look and feel across multiple products. The user interface consists of multiple cooperating processes operating under a layout management system which resides within Host and Host Platform.

The following subsystems contain user interface components which operate within the bounds set by the layout management system.

- Host
- Host Platform
- Viz
- MR Tools

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The user interface layout is defined by PEF within Host Platform. The following diagram show the screen layout enforced by PEF.



The screen header provides single click access to the worklist schedule, image management, and tools (denoted as HLA controls); as well as all active sessions (scanning plus post processing) via the session tables and scanning session tab. The screen header also accommodates the IQE Client UI (when the IQ Engine Option is Active).

The screen footer contains the system status icons as well as quick access to the system messages.

The current scanning session activity is displayed on the right hand side, comprised of the MR Scan Status window, the Auto-View window, and a multi-selection window. The MR Scan Status always shows the status of the current MR Task. The Auto-View displays MR Images as they are reconstructed. The multi-selection can be set to display one of:

- Physio Gating status
- available Coil Element combinations (static or dynamic, see section 4.1.8)
- Protocol Notes

The modality workspace is populated based on the user interface necessary for the operation being performed by the user. For example, when the worklist schedule HLA control is selected, the modality workspace contains the modality worklist user interface.

To ensure windows remain associated with the appropriate HLA or session the PEF layout is augmented by a virtual desktop manager provided within the x-windowing system. The virtual desktop manager allows the Software to define multiple desktops (only one visible at a time) and to constrain windows to exist on either all desktops or on a single desktop.

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The following diagram shows the relationship between the PEF Manager, Desktop Manager and the X Server, FVWM Window Manager and the managed application windows:

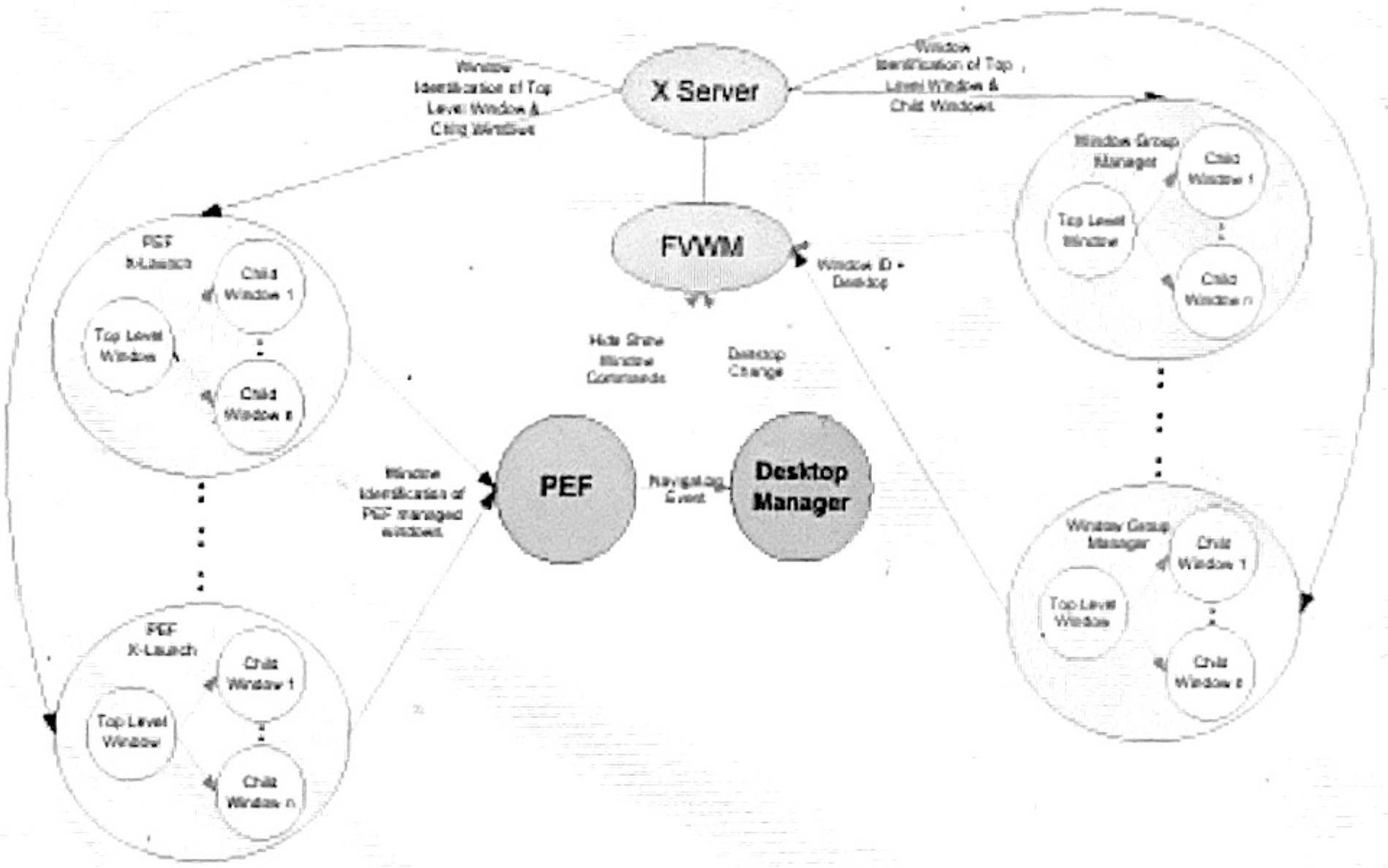


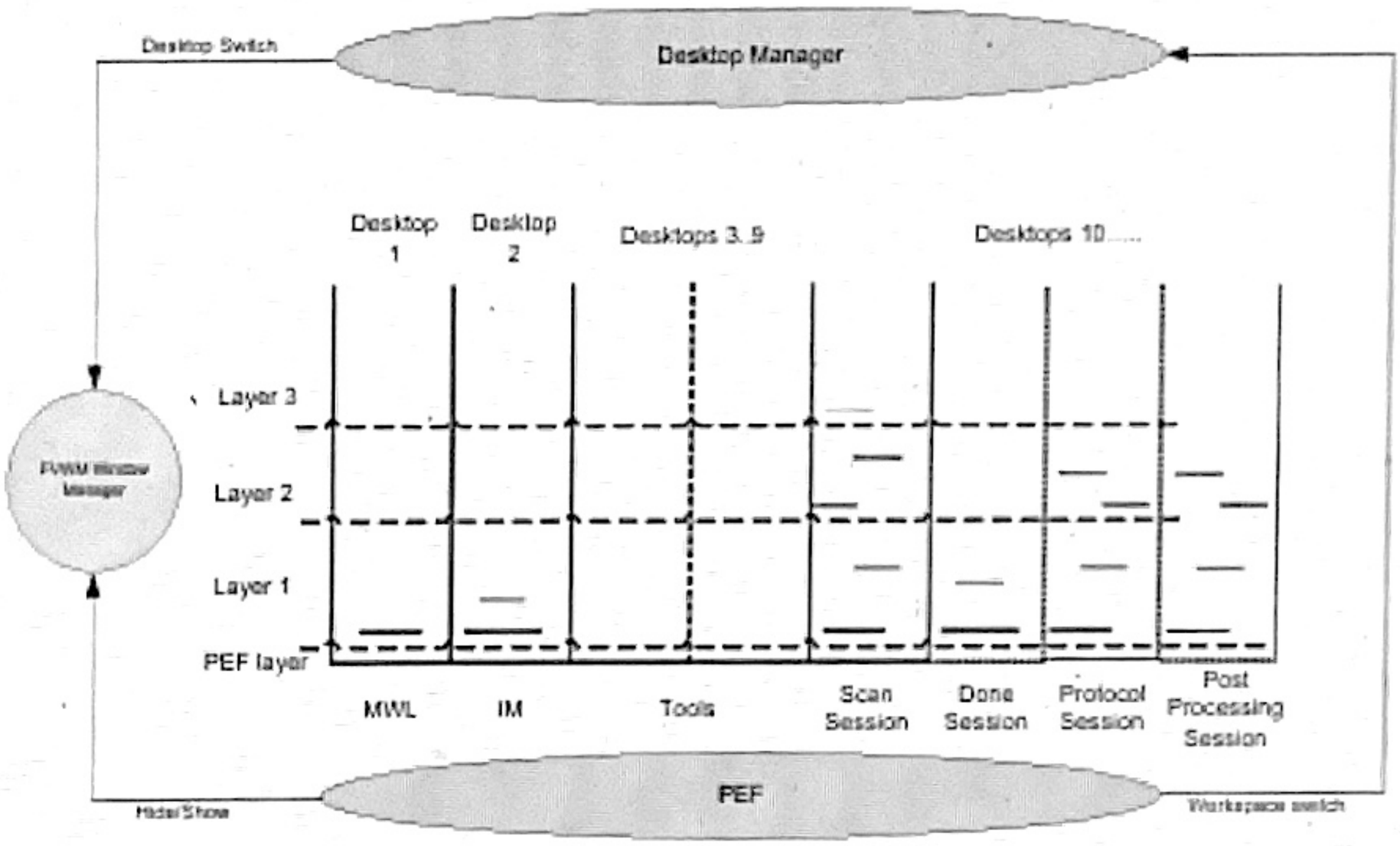
Figure 5-1: Window Management

The active screen layout presents views associated with the HLA navigation (Worklist, Image Management, Tools) as well as the active Sessions (Scan, Scan Done, Post Processing). Within each view the windows are assigned to specific desktops and layers. The layers manage the stacking order for non-PEF managed windows. The PEF managed windows are associated with all desktops, their visibility managed through hide/show commands issued by PEF.

The following diagram shows the relationship between the HLA, sessions, and desktops:

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The specifics of the individual user interfaces exposed by the various subsystems and applications are further described in the detailed design documents associated with the respective components. Refer to Section 3.4.2 for further information.

Additional user interface items consist of the following devices:

- In-Room Operator Console (IROC)
  (available on Premier, some DV, and all PX configurations)
- Touch-and-Go landmark and coil position identification
  (requires IROC)
- Trackballs (2) with selection buttons
  (pointing device for use with IROC)
- Touch Panel
  (pointing device for use with IROC)
- Landmark, Table Motion, Fan, and Lights control buttons
- 7-Segment Display
  (for systems without IROC)

5.2. External Interfaces

The Software provides the following external interfaces.

- DICOM Modality Worklist (MWL)
- DICOM Transfer (Networking)
- DICOM Print
- RSvP (for Remote Service access)
- Secure File Transfer (SFTP)
- Secure Shell (SSH)
- Near real-time send of acquired images to a remote device
- Remote Scanner Control Interface (EXSI)

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- Raw Data Server
- IQ Engine

The external devices and services expected to use the external interfaces are identified in the following table.

Device/Service	External Interface									
	DICOM MWL	DICOM Transfer	DICOM Print	RSvP	SFTP	SSH	RT Send Images	EXSI	Raw Data Server	IQE Client
On-line center				X	X	X				
Workstations										
PACS		X								
DICOM Printer			X							
Scheduling System	X									
Scanner Control Device		X						X		
Special Devices ✖						X	X		X	
Stimulation Devices								X		
IQ Engine										X

✖ For example, the MR touch acoustic device

Security and privacy issues are addressed using network access controls implemented via a software firewall as well as internal access control lists. The software firewall restricts remote access to the scanner to the DICOM interface. The DICOM interface uses an access control list to restrict access to image files to user identified nodes.

5.3. Hardware Interfaces

The primary devices with which the software subsystems interact are listed in the following table.

Hardware	Software Item								
	Host	Host Platform	Viz	MR Recon	PSS	PSI	SCP	Magnet Monitor	Service SW
VRE	X			X					X
Transmit Chain					X*	X	X		X
Receive Chain	X			X	X*	X	X		X
Patient Handling					X**	X	X		X
Gradient Driver					X*	X	X		X
Sequencer					X*	X	X		X
GOC	X	X	X	X	X				X
IROC♣	X	X							
Magnet Sensors								X	

* Via sequencer within PSI and SCP

** Via signal processing unit within PSI and SCP

♣ Optional on DV systems. Mandatory on other systems.

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The hardware interface requirements are covered in the associated SSRS/SWRS documents corresponding to the software components implementing the interface. (Refer to the Requirements columns of the SW Structure [15].)

5.4. Internal Interfaces

Figure 5-2 shows the primary communication paths between the Software subsystems used for scanning activities.

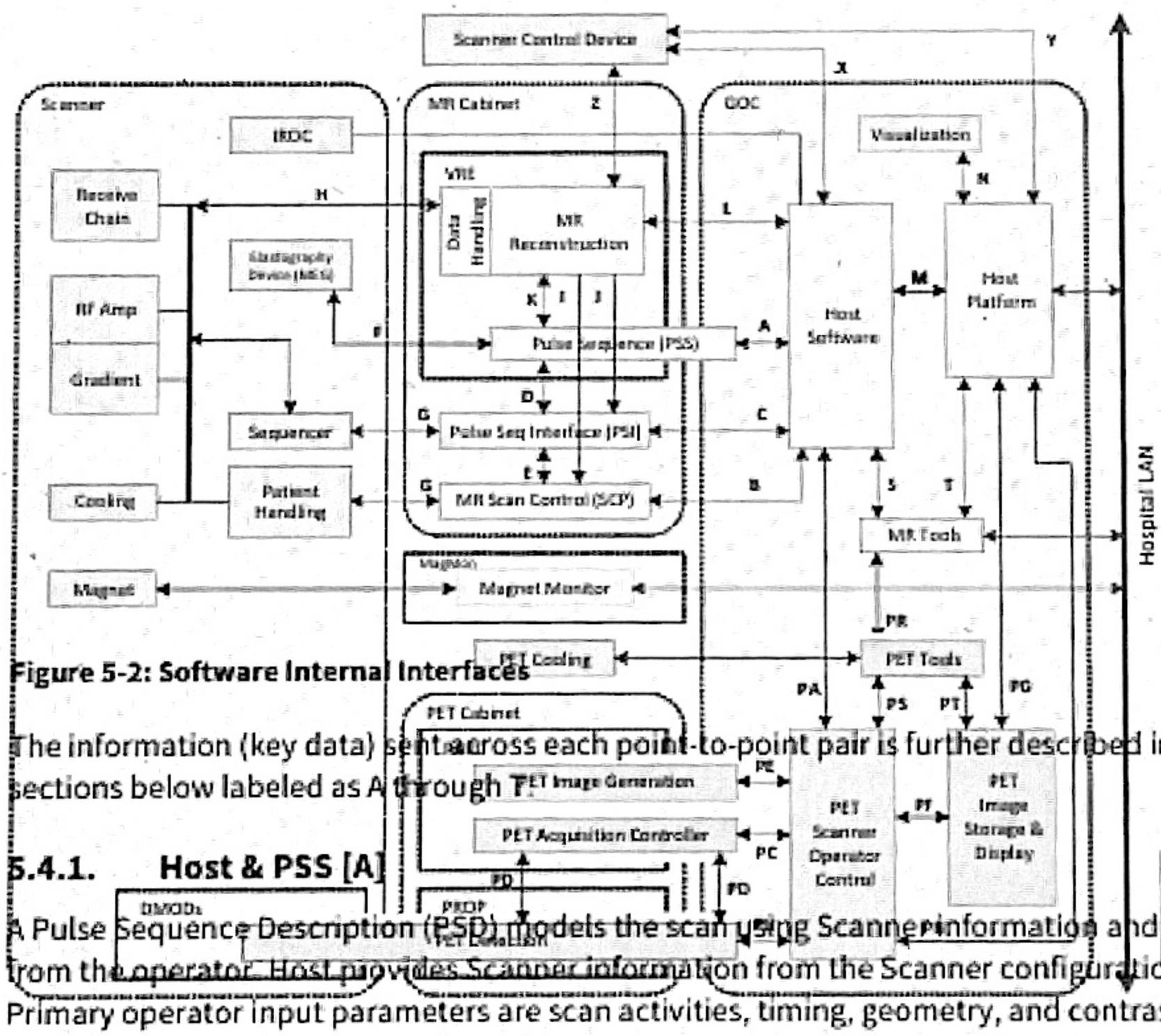


Figure 5-2: Software Internal Interfaces

The information (key data) sent across each point-to-point pair is further described in the sections below labeled as A through T.

5.4.1. Host & PSS [A]

A Pulse Sequence Description (PSD) models the scan using Scanner information and inputs from the operator. Host provides Scanner information from the Scanner configuration. Primary operator input parameters are scan activities, timing, geometry, and contrast. Scan parameters are set into a Host PSD and validated by PSS, which may constrain some parameters due to system and MR physics limitations. The PSD parameters (including any that were constrained) are displayed on the Host UI to guide the operator to valid prescription selections.

The Host PSD also contains information used for reconstruction and in the image header.

The following table summarizes the communications between the subsystems.

From	To	Data	Interface Standard
Host	PSS	user selection	message-Q
PSS	Host	UI/Image/recon information	message-Q

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5.4.2. Host & SCP [B]

Host communicates user actions on the operator console. SCP communicates user actions on hardware.

SCP provides an abstraction of the hardware in the Scanner to Host. The interfaces between Host and SCP are used to communicate configuration information, startup and shutdown commands, prescription selections, execution commands, hardware information, and scan state transitions.

During startup and reset, Host and SCP exchange initialization and configuration information. After startup, SCP sends to Host continuous physiological, table location & status, coil ID information and floating coil stat and position data.

During shutdown, Host sends power off events to SCP and SCP commands the power control hardware to turn off the scanner power after the GOC is shut down.

During scan prescription, SCP sends landmark and physio gating information to Host, and if there are no other applications downloaded, Host loads prescriptions that require longer setup times (such as RF Amp modes) to SCP.

During scan initiation, Host loads the scan prescription to SCP, if the prescription is not already loaded.

During scan execution, SCP handles requests from Host to stop, pause, or resume scanning. Notifications on these events are also sent to Host.

The following table summarizes the communications between the subsystems.

From	To	Data	Interface Standard
Host	SCP	Selected Physio Gating options	MrMail
SCP	Host	Physio Gating Waveform Data	MrMail
SCP	Host	Landmark Position	MrMail
SCP	Host	Connect/Disconnect Coil	MrMail
SCP	Host	Floating Coil State & Position	MrMail
Host	SCP	Table Position & Status	MrMail
Host	SCP	Hardware Settings	MrMail
SCP	Host	Scanner State Start/Stop/Pause	MrMail
Host	SCP	Boot-up Packet	TCP/IP
Host	SCP	Power Off	MrMail

5.4.3. Host & PSI [C]

PSI provides an interface used by Host to download PSDs and associated data. Commands are sent during scan initiation to tell PSI which PSD to download. This also triggers PSI to call into PSS to generate the initial set of waveforms for the scan.

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5.4.4. PSS & PSI [D]

The Target PSD is dynamically loaded (& linked) as an embedded section into by PSI when the operator accepts the modeled scanning activities and instructs the scan to begin. The Host PSD parameterizes the modeled scanning activities and transfers these parameters to the Target PSD through a serialization/de-serialization process called download (see data path A). The Target PSD constructs the waveforms describing the scan using interface functions provided by PSI. The execution of the scan requires dynamic updates to the amplitudes, durations, and relative timing of the pulses describing the scan.

When the scan is initiated, PSI initiates predefined sequence APIs in the Target PSD. These initiation (entry point) APIs invoke PSI APIs that alter the amplitude, control information, geometry, duration, relative timing, and data sorting of the waveforms that compose the sequence. PSS uses APIs provided by PSI to control hardware and check hardware status information.

The following table summarizes the communications between the subsystems.

From	To	Data	Interface Standard
PSS	PSI	waveforms	function calls
PSI	PSS	waveform output state changes	function calls

5.4.5. PSI & SCP [E]

The communication between SCP and PSI mainly occurs during the scan initiation and scan execution phases of operation. PSI is the gateway to PSS and scan play out, which SCP masters.

During scan initiation, SCP obtains additional prescription information from PSI that is not part of the Host prescription information. PSI obtains this from PSS. Scan control also communicates to PSI to make sure the download is complete, before prep can be started.

During scan execution, SCP communicates to PSI to start and stop scan play out for each Pulse Sequence entry point. The APIs that PSI exposes to PSS also communicate to SCP during scan execution to control hardware and check hardware status information.

The following table summarizes the communications between the subsystems.

From	To	Data	Interface Standard
PSI	SCP	Hardware control requests	shared memory & function calls
PSI	SCP	Hardware status requests	shared memory & function calls
SCP	PSI	Hardware status	shared memory & function calls

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5.4.6. PSS & MEG [F]

Pulse Sequence Software (PSS) interacts with the MEG hardware to set run-time parameters on the acoustic driver prior to the start of acquisition.

During the acquisition phase, PSS sends trigger signals to the acoustic driver through the DABOUT6 hardware control.

The following table summarizes of the communications between the subsystems.

From	To	Data	Interface
PSS	HW	Run-time parameters	Standard
HW	PSS	Setup status	UDP
PSS	HW	Trigger Pulse	UDP DABOUT6 digital line

5.4.7. PSI, SCP, & Scanner [G]

PSI and SCP interface to the Scanner through communication links and direct hardware interfaces. These subsystems manage the Scanner, and are responsible for setting up, monitoring the state, and recovering in case of failure.

During scan prescription and scan initiation, as prescription information is received from Host, PSS, and Acquired Data Processing, commands are sent to the Scanner.

During scan execution, real-time changes commanded by PSS through PSI and SCP are translated into hardware changes, and sent to the scanner hardware.

The following table summarizes the communications between the subsystems.

From	To	Data	Interface
PSI & SCP	HW	Hardware state changes	Standard memory mapped IO
PSI & SCP	HW	Hardware state changes	messages over comm. link
HW	PSI & SCP	Hardware status	memory mapped IO
HW	PSI & SCP	Hardware status	messages over comm. link

5.4.8. Receive Chain & MR Recon [H]

The receive chain hardware samples the MR signal simultaneously from the entire set of receiver antennas that are embedded in the active receive coil. The sampling is initiated at specific points in the course of scanning, controlled by PSD. MR Recon configures the receive chain hardware with the number of points to sample during scan prep. Each sampling window delivers a "frame" of raw MR data to MR Recon. An acquired frame consists of:

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